

ARRAYS IN JAVA

A Powerful Way to Handle Collections of Data

INTRODUCTION TO ARRAYS

- An array is a data structure used to store multiple values of the same type.
- Each element in an array is stored at a specific index
- Index starts from 0.

WHY USE ARRAYS?

- Groups multiple values under one variable name.
- Easy to manage and process with loops.
- Improves performance when handling large datasets.

ARRAY DECLARATION & INITIALIZATION

LOGIC:

// Declaration

```
int[] arr;
```

// Initialization

```
arr = new int[5];
```

// Combined

```
int[] arr = {10, 20, 30, 40, 50};
```


LOOPING THROUGH ARRAYS

```
for (int i = 0; i < arr.length; i++)  
    System.out.println(arr[i]);
```

```
for (int num : arr)  
    System.out.println(num);
```

- `arr.length` gives the array size.

TYPES OF ARRAYS

- One-dimensional arrays
- Two-dimensional arrays
- Multi-dimensional arrays

2D ARRAYS IN JAVA:

- A 2D array is an array of arrays, storing data in rows and columns.
- It is declared as `int[][] arr = new int[rows][cols];`.
- Elements are accessed using two indices like `arr[0][1]`.

2D ARRAY EXAMPLE

```
int[][] matrix = {  
    {1, 2, 3},  
    {4, 5, 6}  
};
```

```
System.out.println(matrix[0][1]); // prints 2
```


ADVANTAGES AND LIMITATIONS

✓ Advantages:

- Fast access via index
- Simple and memory-efficient

✗ Limitations:

- Fixed size
- Only same type elements
- No built-in methods like ArrayList

The background features a light gray field with a series of thin, white, wavy lines that create a sense of movement. In the corners, there are decorative elements: the top-left and bottom-right corners have thick, flowing bands of dark blue and gold, while the top-right and bottom-left corners feature a subtle, thin-lined grid pattern.

THANK YOU